

Proposed Title: *AI-Driven UAV Building Inspection*

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1. Book Overview / Synopsis

Background and Motivation

The aging of the urban building stock and the rapid expansion of modern cities have made building inspection and structural health monitoring (SHM) more critical than ever. Traditional manual inspection methods are notoriously labor-intensive, time-consuming, and pose significant safety risks to human inspectors. In recent years, Unmanned Aerial Vehicles (UAVs) have emerged as a transformative technology in this domain, offering a safer and more efficient alternative for data collection. However, the current literature often treats the UAV inspection process in a fragmented manner—focusing either purely on drone control or solely on computer vision. There is a pressing need for a comprehensive resource that bridges the gap between mechanical data acquisition and high-level, AI-driven structural analysis.

The Book's Core Contribution

AI-Driven UAV Building Inspection proposes a holistic, end-to-end framework for automating the entire building inspection pipeline. This multi-authored monograph brings together leading experts across robotics, computer vision, spatial data management, and artificial intelligence to present a unified approach to structural monitoring. By seamlessly integrating the physical operation of UAVs with cutting-edge digital analysis, this book provides readers with a complete roadmap for deploying intelligent inspection systems.

Detailed Thematic Breakdown

The book is strategically structured into four interconnected phases, guiding the reader from the physical flight of the drone to the generation of intelligent, actionable reports:

- 1. Intelligent Data Acquisition (Task & Motion Planning):** The journey begins with the physical deployment of UAVs. The book details advanced **Task Planning** methodologies, addressing how to efficiently allocate resources, manage sensor payloads, and design coverage paths for complex architectural structures. Following this, it delves into **Motion Planning**, exploring real-time obstacle avoidance, dynamic trajectory optimization, and safe navigation in unpredictable urban environments. This ensures that the data collected is both comprehensive and captured safely.
- 2. Digital Representation and Spatial Context (3D Reconstruction & GIS):** Once high-quality visual data is captured, the book explores state-of-the-art **3D Reconstruction** techniques. It covers the transition from raw 2D images and LiDAR point clouds to high-fidelity 3D models and Building Information Models (BIM). To manage these models on a macro scale, the text introduces **Geographic Information Systems (GIS)** integration, enabling inspectors to map UAV data spatially, conduct spatiotemporal degradation analysis, and manage large-scale city assets within a Digital Twin framework.
- 3. AI-Driven Defect Analysis (Datasets, Detection & Evaluation):** At the core of the inspection process is the identification of structural anomalies. The book provides a rigorous guide to **Dataset Processing and Construction**, detailing how to curate, annotate, and augment data for structural defects (e.g., cracks, spalling). It then evaluates the latest deep learning models for precise **Defect Detection and Segmentation**, offering robust evaluation metrics to ensure the reliability and accuracy of the AI models in real-world scenarios.

4. **Next-Generation Decision Support (Large Language Models):** Pushing the boundaries of current inspection paradigms, the final section introduces the application of **Large Language Models (LLMs)** and Vision-Language Models (VLMs) in civil engineering. It investigates how generative AI can be fused with visual defect data to automatically generate comprehensive inspection reports, query historical maintenance records naturally, and provide intelligent decision support and predictive maintenance recommendations.

2. Impact and Target Audience

By synthesizing these diverse yet complementary disciplines—UAV control, 3D modeling, deep learning, GIS, and generative AI—this book stands as a pioneering reference. It is designed to serve as an essential resource for academic researchers, graduate students, civil engineers, and technology developers who are looking to build, deploy, or study the next generation of autonomous structural inspection systems.

3. Proposed Table of Contents

Chapter 1: Introduction to UAV-Based Building Inspection

- 1.1 The Evolution of Structural Health Monitoring
- 1.2 Advantages of UAVs in Building Inspection
- 1.3 Overview of the Automated Inspection Pipeline
- 1.4 Structure of the Book

Chapter 2: UAV Task Planning for Data Collection (Lead: Xinyi)

- 2.1 Introduction to Task Planning in Aerial Inspection
- 2.2 Multi-UAV Coordination and Task Allocation
- 2.3 Sensor Selection and Payload Management
- 2.4 Coverage Path Planning for Complex Building Surfaces
- 2.5 Energy-Aware Optimization in Task Execution
- 2.6 Case Studies: Task Planning for Large-Scale Buildings

Chapter 3: UAV Motion Planning in Complex Environments (Lead: Chuanxiang)

- 3.1 Fundamentals of UAV Navigation systems
- 3.2 Real time mapping and localization methods
- 3.3 Obstacle-aware Trajectory Generation for complex environment
- 3.4 Energy-efficient Trajectory Generation for large-scale environment
- 3.5 Simulation and Real-World Flight Experiments

Chapter 4: 3D Reconstruction of Buildings (Lead: Guidong)

- 4.1 Role of 3D Reconstruction in UAV-Based Building Inspection
- 4.2 Structure-from-Motion for Sparse Scene Reconstruction
- 4.3 Geometry-Based Dense Reconstruction Methods
- 4.4 Neural Rendering-Based Reconstruction Methods
- 4.5 Point Cloud Registration and Surface Reconstruction
- 4.6 Reconstruction Quality Assessment and Inspection-Oriented Evaluation

Chapter 5: Dataset Construction and Benchmarks (Lead: Benyun)

- 5.1 The Role of Datasets in UAV Façade Inspection
- 5.2 Image Preprocessing and Annotation
- 5.3 A Large-Scale Benchmark for Façade Defect Detection

- 5.4 Extending the Benchmark to Instance Segmentation
- 5.5 Evaluation of Detection and Segmentation Models

Chapter 6: AI Models for Building Inspection (Lead: Benyun)

- 6.1 Real-Time Constraints on UAV Platforms: Compute, Power, and Latency
- 6.2 Architectural Foundations
- 6.3 AI Models for Defect Detection
- 6.4 AI Models for Defect Instance Segmentation
- 6.5 Comparative Analysis and Architectural Insights
- 6.6 Edge Deployment and On-Board Field Validation

Chapter 7: GIS Integration for Building Monitoring (Lead: Jihan)

- 7.1 Introduction to Geographic Information Systems (GIS)
- 7.2 Mapping UAV Data and 3D Models to GIS Platforms
- 7.3 Spatiotemporal Analysis of Structural Degradation
- 7.4 Bridging GIS and Building Surface Models
- 7.5 Large-Scale City Inspection and Asset Management
- 7.6 Visualization and Interactive Dashboards

Chapter 8: Digital Twin with LLMs in Inspection Analysis (Lead: Yijun)

- 8.1 Digital Twins for Building Inspection
- 8.2 Foundations of LLMs for Civil Engineering Applications
- 8.3 From Digital Twins to Language Reasoning
- 8.4 Multi-Modal Fusion
- 8.5 Inspection Report Generation and Management
- 8.6 Decision Support and Predictive Maintenance

Chapter 9: Conclusion and Future Prospects

- 9.1 Summary of the Integrated UAV Inspection Framework
- 9.2 Regulatory, Privacy, and Safety Considerations
- 9.3 The Future: Fully Autonomous Swarms and Next-Gen AI Integration

4. Timeline: 6 months

5. Market Competition:

While there are several existing titles on drone applications in civil engineering (e.g., *Unmanned Aerial Vehicles in Civil Engineering* or basic guides on UAV photogrammetry), the current market literature remains heavily siloed. Existing books focus almost exclusively on either the mechanical control and navigation of drones or the fundamentals of computer vision.

This monograph distinguishes itself by offering a truly end-to-end, AI-native pipeline. It is uniquely positioned as the first comprehensive textbook to integrate Large Language Models (LLMs), and Digital Twins with traditional UAV data collection and GIS. By bridging the gap between raw spatial data acquisition and generative AI-driven decision support, this book provides a next-generation roadmap that currently does not exist in the academic or professional market.